

☐ د ٤

☐ هـ ٤

6

* رقم الكمبيوتر

2

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

DJQX

 Will be reviewed

8

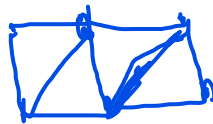
How many triangles are drawn if we use the drawing mode `GL_TRIANGLE_STRIP` with six vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`?

☐ 2

☐ 3

☒ 4

☐ 5



 Will be reviewed

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glPushMatrix();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glScalef (-1.0,0.5, -1.0);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

10

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glScalef(-1.0, 2.0, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☒ B

☐ C

☐ D

 Will be reviewed

11

This 3D homogeneous point: (4,3,2,1) corresponds to what Cartesian point:

☐ (4,3,2,1)

☐ (3,2,1)

☐ (3/4, 1/2, 1/4)

☒ (4,3,2)

 Will be reviewed

12

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW); glLoadIdentity(); glRotatef(45.0, 0, 0, 1.0);  
glRotatef(45.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

13

The following two code segments are equivalent:

Code#1

```
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
drawShape_3();
```

Code#2

```
glRotatef(-90.0, 0, 0, 1.0);  
glTranslatef(-1.0f, -2.0f, -3.0f);  
drawShape_3();
```

☐ True

☒ False

 **Will be reviewed**

14

Generally, what primitive polygon is used for creating a mesh to represent a complex object?

☒ Triangle

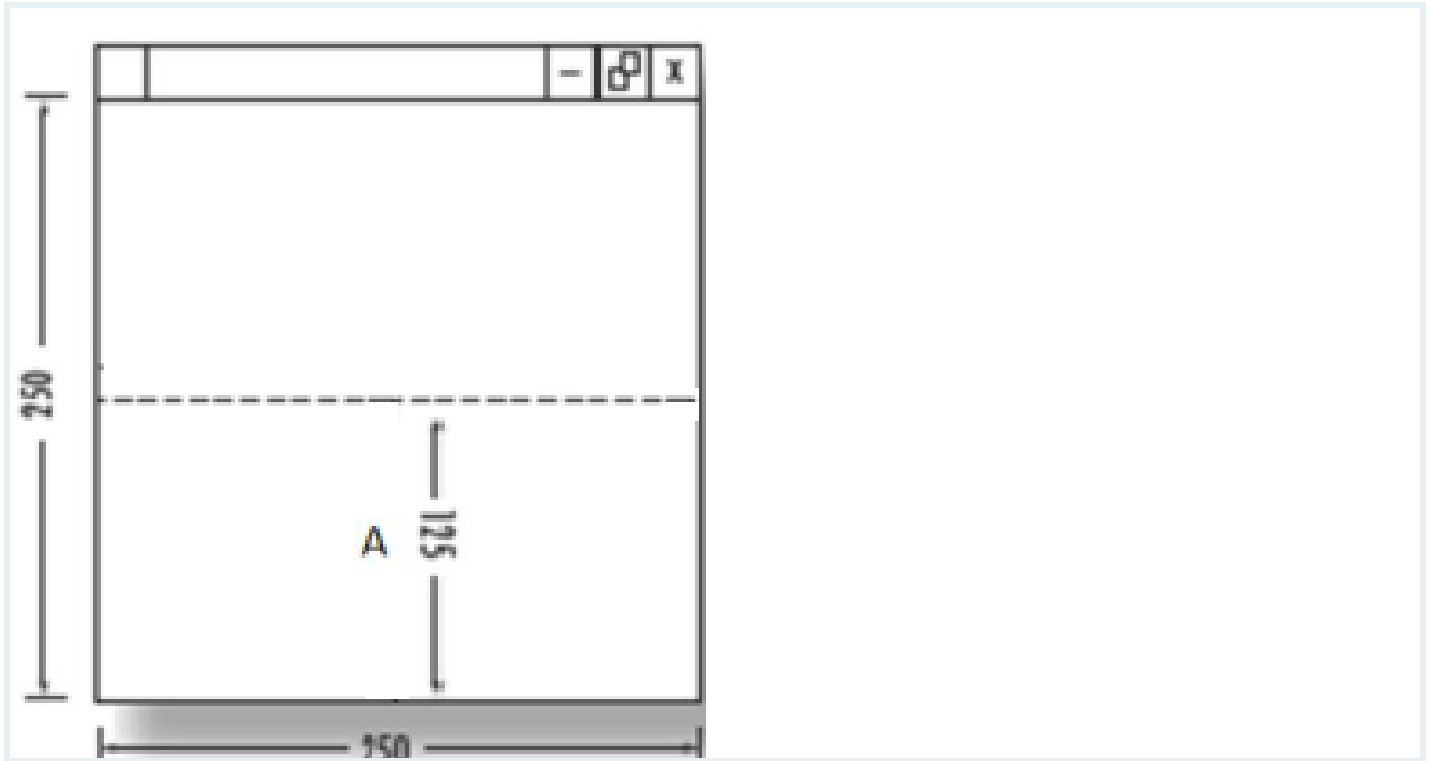
☐ Cube

☐ Sphere

☐ Circle

 **Will be reviewed**

To map the output of the OpenGL code to the area A (as shown in the figure), we use the.



- ☒ `glViewport(0, 0, 250, 125);`
- ☐ `glViewport(0, 0, 125, 250);`
- ☐ `glViewport(125, 125, 125, 125);`
- ☐ `glViewport(250, 125, 0, 0);`
- ☐ `glViewport(0, 250, 0, 125);`

 **Will be reviewed**

OpenGL can draw convex polygons.

- ☒ True

☐ False

 Will be reviewed

17

Rotate (z, 90) * Rotate (z, 45) = Rotate (z, 45) * Rotate (z, 90)

☒ True

☐ False

 Will be reviewed

18

The OpenGL represents the following matrix as

$$m = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☐ GLfloat m[] = { 1.0f, 5.0f, 1.0f, 2.0f, 6.0f, 2.0f, 3.0f, 7.0f, 3.0f, 4.0f, 8.0f, 4.0f};

☐ GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f};

☐ GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f, 0.0f, 0.0f, 0.0f, 1.0f};

☒ GLfloat m[] = { 1.0f, 5.0f, 1.0f, 0.0f, 2.0f, 6.0f, 2.0f, 0.0f, 3.0f, 7.0f, 3.0f, 0.0f, 4.0f, 8.0f, 4.0f, 1.0f};

 Will be reviewed

19

What does glClear do?

- ☒ It clears all OpenGL buffers
- ☐ It clears the OpenGL buffers specified in an argument
- ☐ It sets the window color to black
- ☐ None of the above

 **Will be reviewed**

20

Which of the following is **not** a valid argument to glBegin ()?

- ☐ GL_POINTS
- ☐ GL_LINES
- ☐ GL_LINE_STRIP
- ☐ GL_TRIANGLES
- ☒ GL_QUADS_FAN

 **Will be reviewed**

21

glOrtho is used to define a _____ matrix.

- ☐ modelview
- ☒ projection

- ☐ view
- ☐ model

📝 Will be reviewed

22

What is the significance of i,f,v,d... in function naming conventions found in OpenGL?

- ☐ Specify the target type for variables written by OpenGL on the CPU-side only
- ☐ OpenGL's naming conventions vary and must be examined in the specification for the version you are using
- ☒ Specify the expected parameter types (int, float, pointer, double, etc.)
- ☐ Specify the output target (immediate mode, frame buffer, virtual mode, double-buffer, etc.)

📝 Will be reviewed

23

Rotating the 3D point (0,-1,0) by -90 around z-axis gives what:

- ☐ (1,0,0)
- ☒ (-1,0,0)
- ☐ (0,1,0)
- ☐ (0,-1,0)
- ☐ (-1,-1,0)

 Will be reviewed

24

Consider the following matrix.

This matrix achieves the following in the global coordinate system: (Note: R_z means rotation around the z axis and S_z means scale in the z direction)

0.707	0.707	0	0
-0.707	0.707	0	0
0	0	2	0
0	0	0	1

- ☐ R_z of 45° followed by S_z of 2
- ☐ R_y of 45° followed by S_z of 2
- ☒ S_z of 2 followed by R_z of -45°
- ☐ S_z of 2 followed by R_y of 45°

 Will be reviewed

25

The plane N is given by $2x+2y+2z = 6$ What is the normal to the plane N ?

- ☐ $(1, 1, 0)$
- ☒ $(1, 1, 1)$
- ☐ $(0, 1, 0)$
- ☐ $(1, 0, 1)$

 Will be reviewed

Is the following code correct?

```
glBegin (GL_QUADS);  
glVertex2i (1, 0);  
glVertex2i (1, 1);  
glVertex2i (-1, 1);  
glVertex2i (-1, 0);  
glEnd (GL_QUADS);
```

☐ Yes

☒ No

 **Will be reviewed**

Which of the following is not a valid argument to glMatrixMode?

☐ GL_PROJECTION

☒ GL_DEPTH

☐ GL_MODELVIEW



This content is created by the owner of the form. The data you submit will be sent to the form owner. Microsoft is not responsible for the privacy or security practices of its customers, including those of this form owner. Never give out your password.

Microsoft Forms | AI-Powered surveys, quizzes and polls [Create my own form](#)

[Privacy and cookies](#) | [Terms of use](#)